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ASSESSMENT COVER SHEET

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Unit	Data and Visualisation for Business
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Assessment	2
Assessment Task	Data Visualisation - Modelling Task
Report Title	A Visual Analysis of NSW Public School Enrolments and Distribution

1. Introduction

Data visualisation translates complex datasets into accessible visual narratives that support evidence-based decision making (Cairo, 2019). This report analyses the Master dataset: NSW government school locations and student enrolment numbers published by the NSW Department of Education (2026), which records the location, type, enrolment and demographic attributes of every government school in New South Wales. Five visualisations were produced in R using the `ggplot2` grammar of graphics (Wickham, 2016) to examine the size, distribution, remoteness profile and historical growth of the network. The report documents the data preparation, presents and interprets each visualisation with its design rationale, and reflects on the ethical principles guiding the work.

2. Data Preparation, Transformation and Cleaning

The raw CSV was imported with `readr` and inspected using `summary()` to establish each variable's type, range and completeness. Categorical fields, including level of schooling, remoteness classification, gender, electorate and local government areas, were converted to factors so they could be grouped and mapped to colour aesthetics, while the date fields were parsed as dates to enable time-series analysis. Enrolment is reported in full-time-equivalent (FTE) units to account for part-time study.

Missing and suppressed values were then addressed. The empty `Support_classes` column was removed. The `Indigenous_pct` and `LBOTE_pct` fields contained the placeholder “np”, which the custodian uses to suppress figures wherever five or fewer students are involved, protecting privacy (NSW Department of Education, 2026). Coercing them to numeric converted “np” to NA, after which numeric columns were imputed with their median, a robust measure given the right-skew in enrolment data. Rows with missing categorical values were dropped and the data checked for duplicates.

Finally, the remoteness classification was also reordered into a logical urban-to-remote sequence rather than the default alphabetical order, so the comparative charts read correctly along their natural gradient. The cleaned dataset was then exported to a CSV file.

3. Visualisations, Analysis and Rationale

3.1 Distribution of Student Enrolments

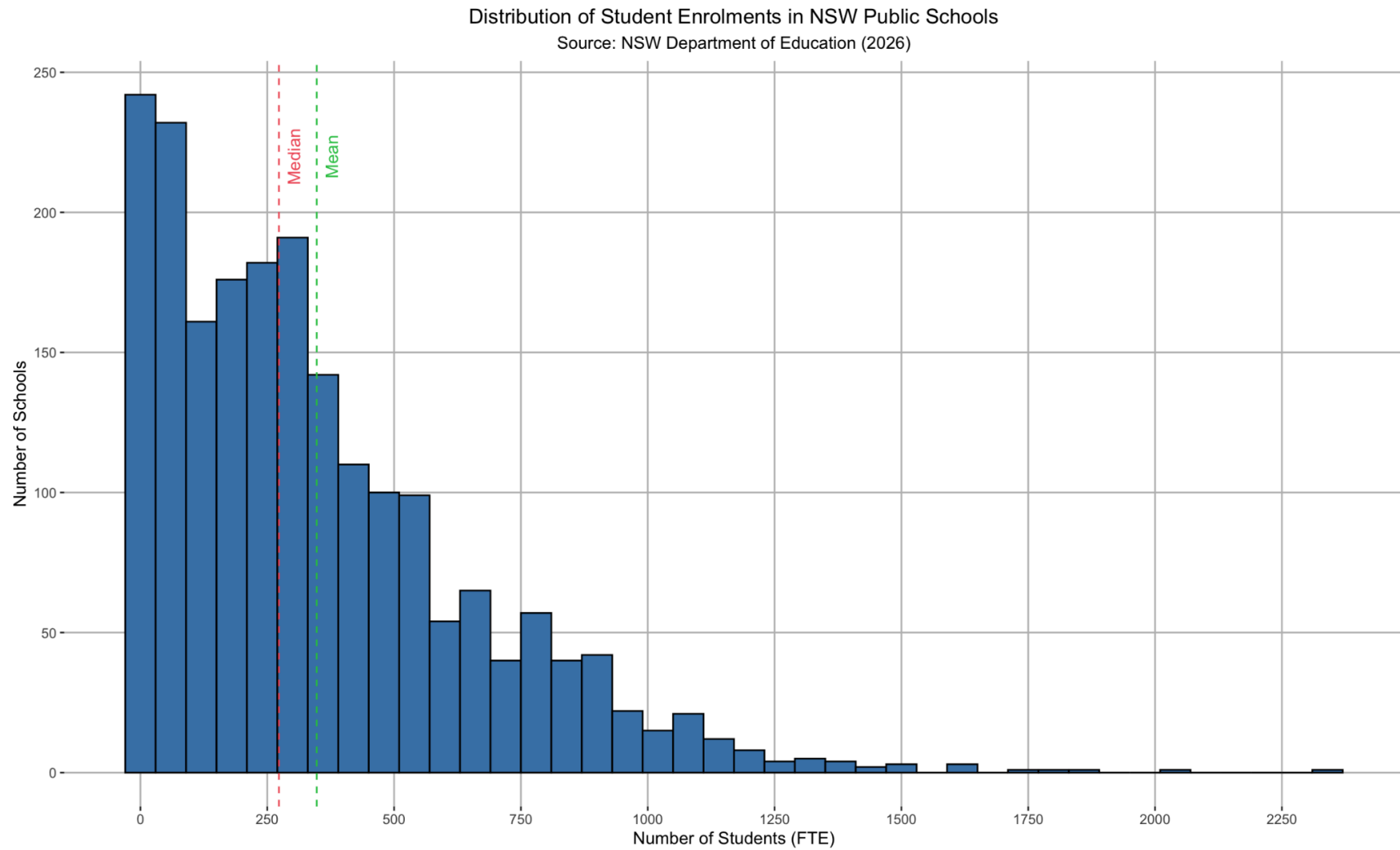


Figure 1. Histogram of student enrolments (FTE) across NSW public schools

A histogram is the standard tool for revealing the shape, centre and spread of a continuous variable (Wickham, 2016); the blank panel keeps attention on the data rather than clutter (Tufte, 2001). The distribution is strongly right-skewed: most schools enrol fewer than 400 FTE students, while a long, thin tail extends beyond 2,000, so the median best summarises a typical school. Because small schools so heavily outnumber large ones, departmental reporting and funding formulas should be benchmarked against the median rather than the mean, and a per-school fixed-cost funding floor should be considered so that the many sub-400-FTE schools are not disadvantaged by allocation driven purely by enrolment.

3.2 Geographic Distribution by Level and Size

Distribution of Public NSW Schools: Level of Schooling and Student Population

Source: NSW Department of Education (2026)

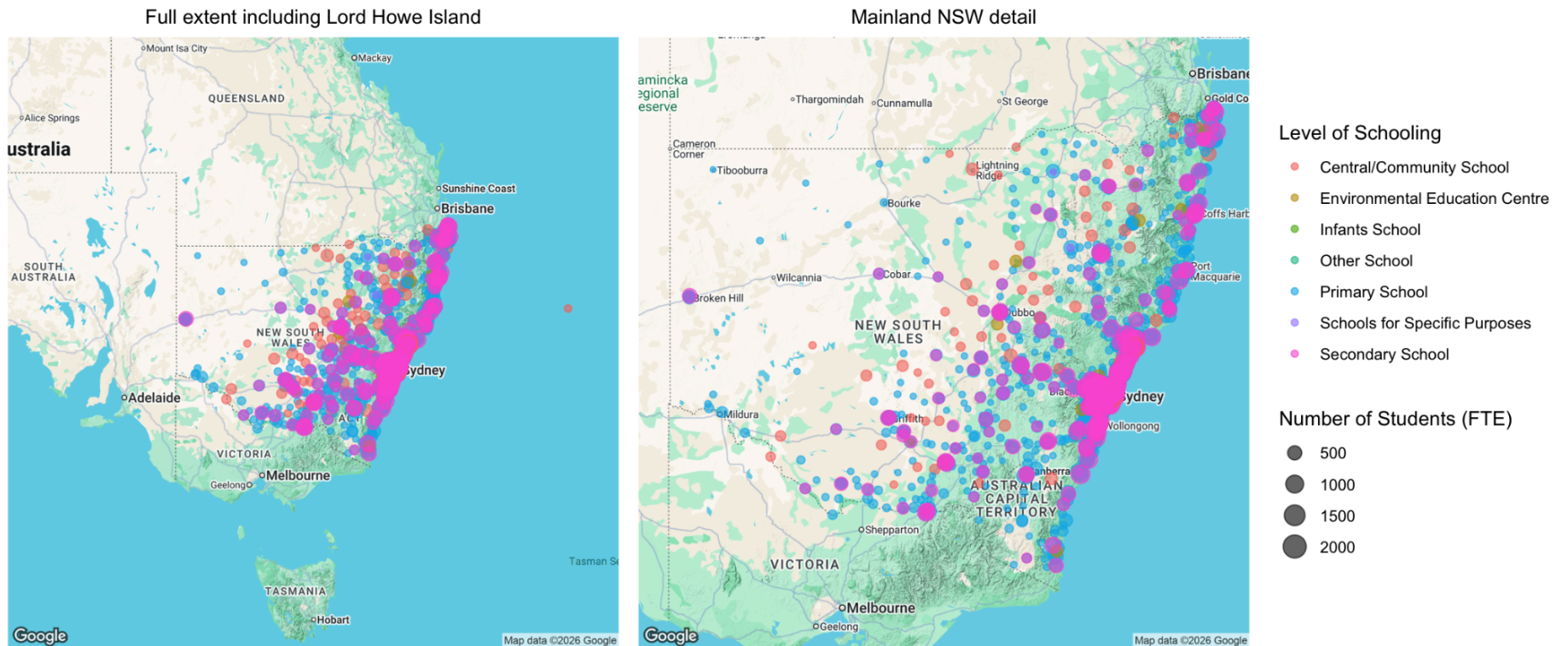


Figure 2. Dual-panel point map of schools by level of schooling (colour) and enrolment (point size).

Two point maps were placed side by side using ggmap and patchwork. A map is the natural encoding for spatial coordinates, with colour for school level and point size for enrolment. The dual-panel design resolves a trade-off: the full panel retains Lord Howe Island offshore, while the mainland panel resolves the dense coastal corridor the island would otherwise compress. Schools cluster densely along the eastern coast and around Sydney, Newcastle and Wollongong, where the largest secondary campuses (the biggest points) are concentrated, while the arid western interior carries only a thin scatter of small, mostly primary schools. The visible service gap across the western interior suggests prioritising distance-education capacity, transport subsidies and digital connectivity for these under-served inland communities over the already well-provisioned coast.

3.3 School Levels Across Remoteness Areas

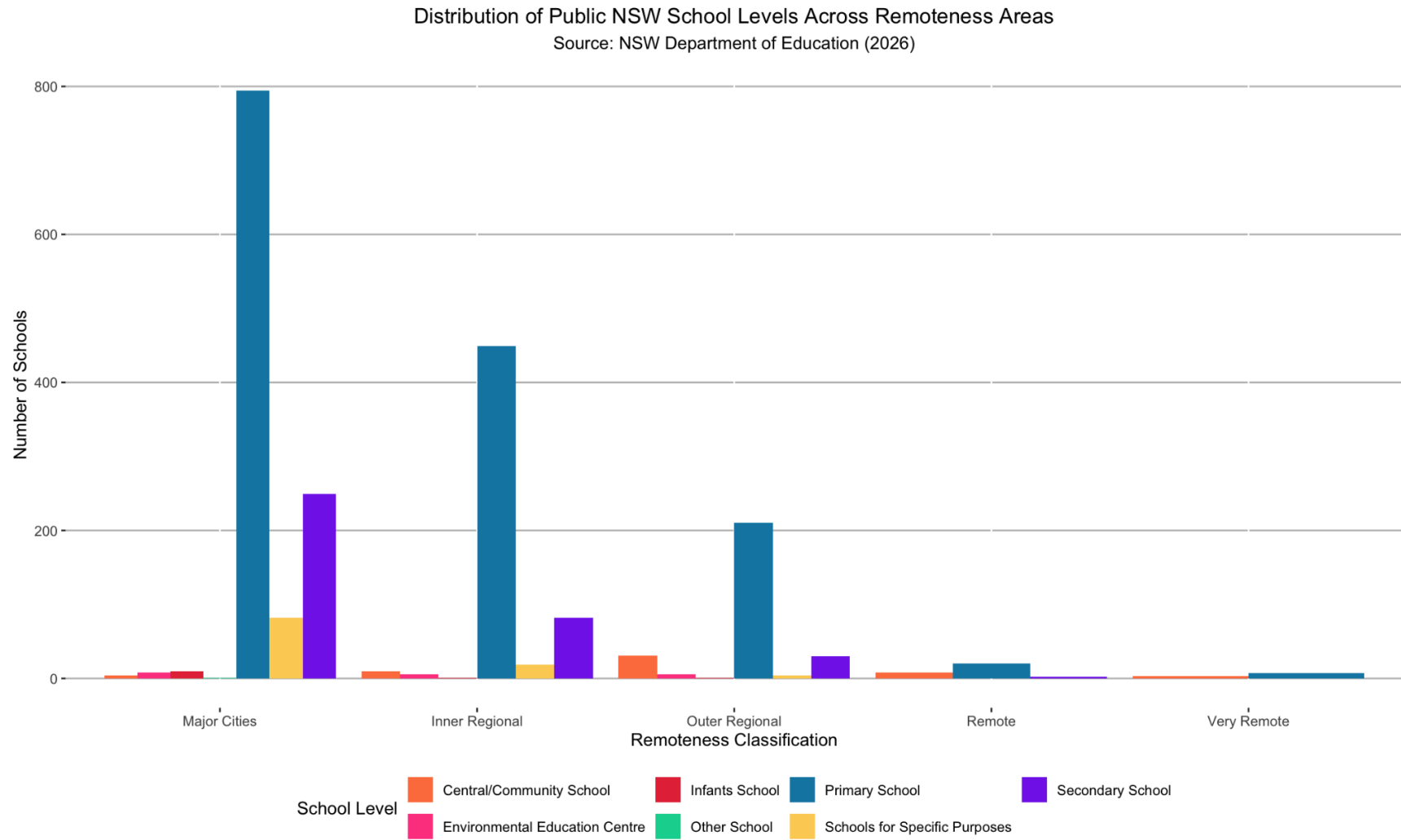


Figure 3. Grouped bar chart of school level counts within each remoteness class.

A grouped (dodged) bar chart compares the count of each school level within the five Australian Statistical Geography Standard remoteness classes (Australian Bureau of Statistics, 2021). Dodging rather than stacking gives every bar a common baseline, essential for accurate length comparison (Cairo, 2019). Major Cities hold by far the most schools, close to 800 primary alone, and the count falls steeply with remoteness, leaving Very Remote areas with only a handful. Primary schools dominate every class, while secondary schools and schools for specific purposes are concentrated in Major Cities and become almost absent further out. The Department should therefore expand distance-education secondary pathways and mobile or itinerant specialist services so that students in remote and very remote areas can access senior and specialist schooling without being required to relocate.

3.4 School Size by Remoteness Classification

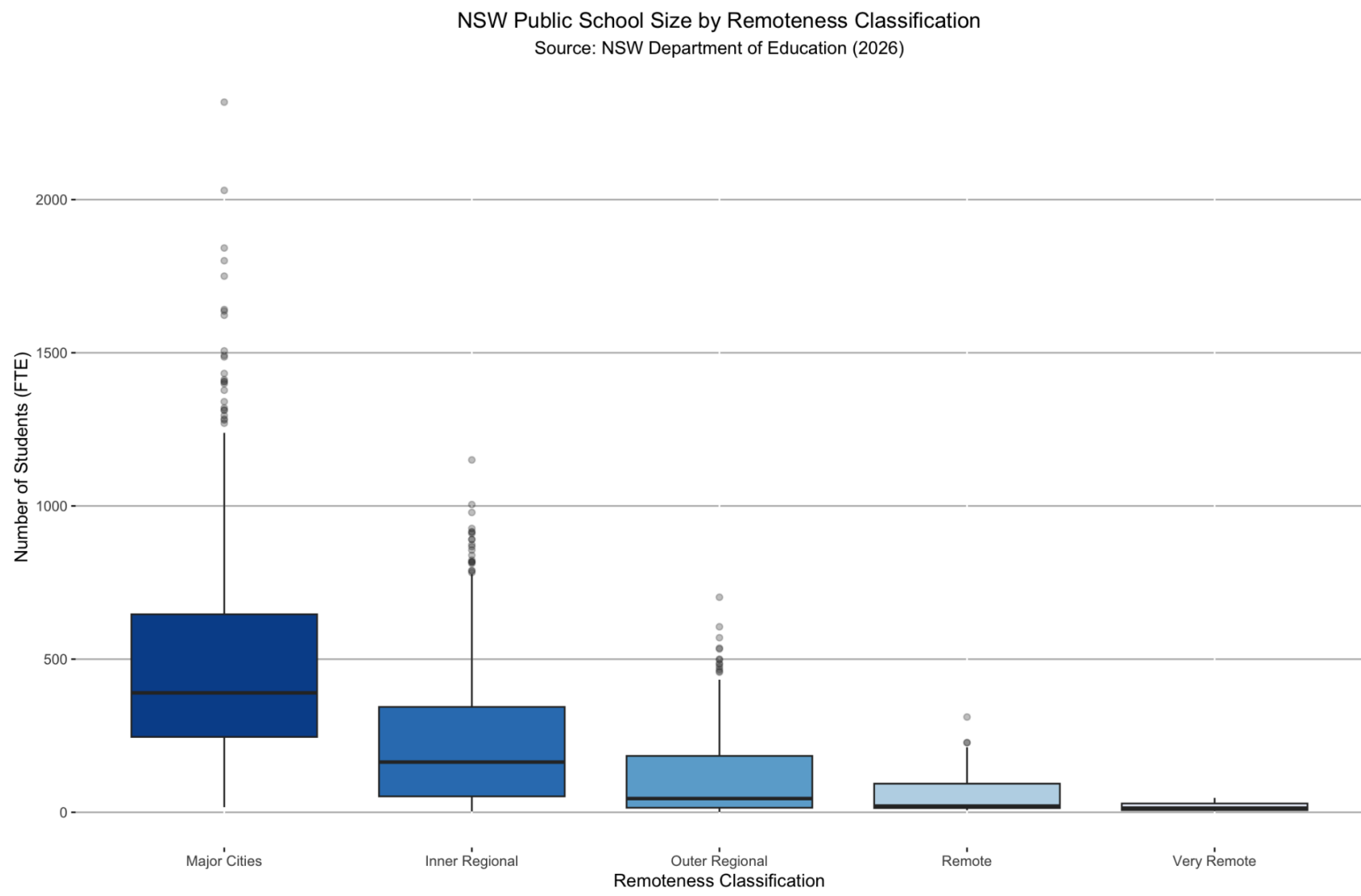


Figure 4. Box plots of enrolment (FTE) by remoteness class.

A box plot compares the enrolment distribution across remoteness classes, summarising median, spread and outliers compactly. Median school size declines monotonically with remoteness: Major Cities have the highest median (around 390 FTE), the widest spread and the most extreme outliers, several exceeding 2,000 FTE, while Very Remote schools are tiny. Given this consistent gradient, staffing and funding for remote schools should be weighted rather than scaled purely on enrolment, for example through guaranteed minimum staffing, multi-grade teaching support and rural retention incentives, since enrolment-based allocation would systematically underfund the smallest, most isolated schools.

3.5 Growth of the School Network Over Time

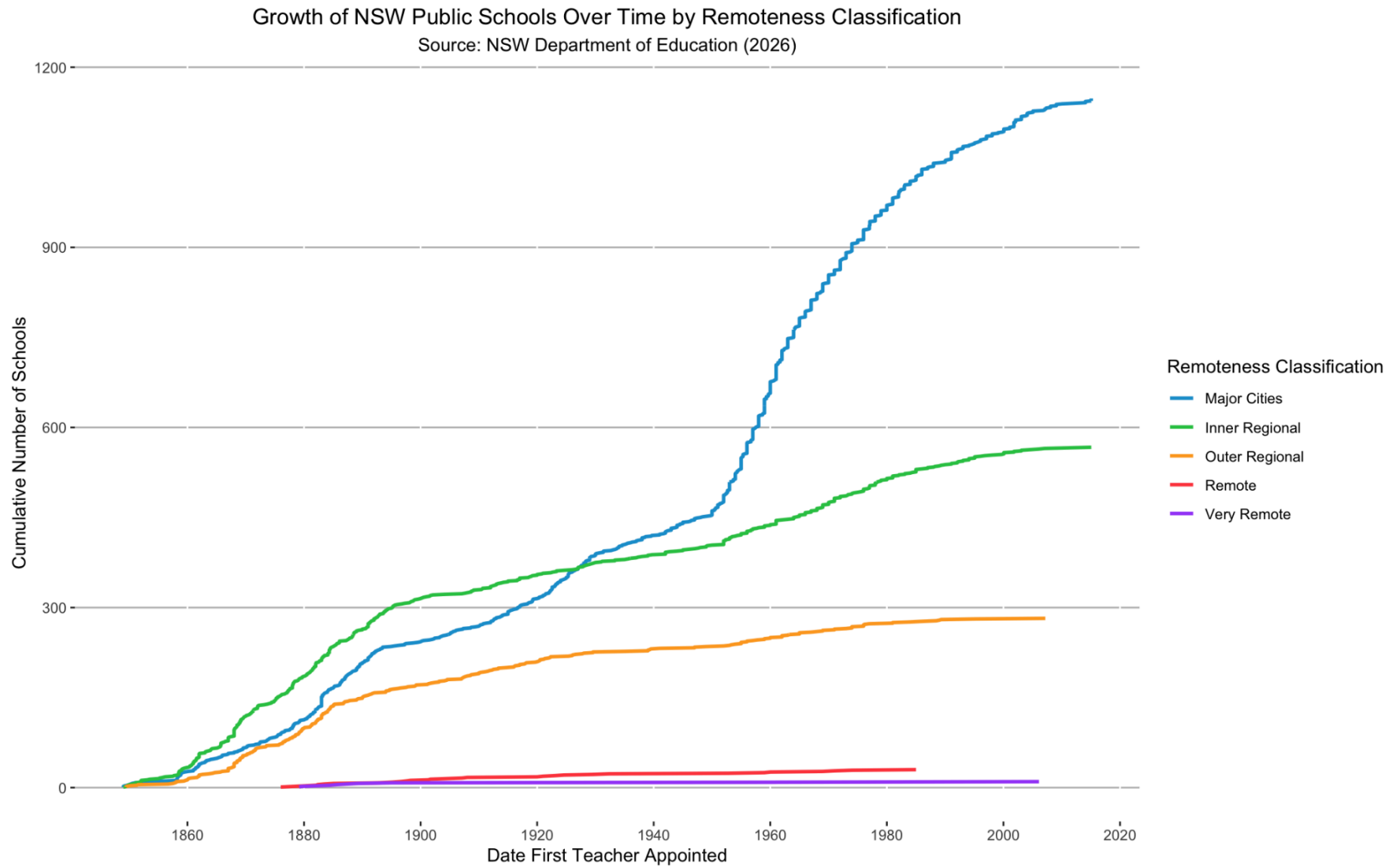


Figure 5. Cumulative number of schools by date of first teacher appointment, grouped by remoteness.

A cumulative line chart plots the running count of schools against the date their first teacher was appointed, grouped by remoteness. A line suits continuous time-series data, and the cumulative form foregrounds long-run accumulation. Inner regional schools were established slightly ahead of metropolitan ones through the nineteenth century, but Major Cities overtook them around the 1920s and accelerated dramatically from mid-century, roughly doubling the inner regional total. The remote and very remote curves stay almost flat throughout. These flat curves warrant monitoring to confirm provision keeps pace with population change; the historical appointment-date field should also be cleaned and validated before use in longitudinal planning.

4. Interpretation: Patterns, Trends and Anomalies

Read together, the five visualisations describe a network governed primarily by population geography. The clearest pattern is a consistent remoteness gradient: as remoteness increases, both the number of schools (Figure 3) and their typical size (Figure 4) fall in step, and the coastal concentration of large campuses (Figure 2) expresses the same effect. Major Cities lead decisively, holding both the most schools and the largest. The right-skewed size distribution in Figure 1 is therefore structural, many small schools serving dispersed communities alongside a few large schools in dense urban catchments.

The dominant trend is historical. The cumulative growth curves in Figure 5 reveal a reversal: inner regional areas were settled and schooled first, leading the metropolitan network through the nineteenth century, before the Major Cities curve overtook it around the 1920s and surged through the mid-twentieth century. This acceleration aligns with the post-war suburban expansion of Sydney, driven by the baby boom and immigration. Remote and very remote growth, by contrast, has been negligible throughout, reinforcing that low population rather than recent policy sets the ceiling on rural provision.

Several anomalies merit comment. The most extreme upper outliers in Figure 4 sit in Major Cities, with some schools exceeding 2,000 FTE; although statistical outliers, these are genuine large metropolitan secondary colleges, consistent with dense urban catchments. Lord Howe Island appears as a striking geographic outlier in Figure 2, yet verification confirmed it as a legitimate central school rather than a coordinate error and it was retained. Finally, the clustering of very early teacher-appointment dates in Figure 5 most plausibly reflects record-keeping conventions rather than a genuine surge of foundations, and is a data-quality limitation.

These patterns share a few contributing factors: the distribution of the State's population, the minimum enrolment to sustain a secondary or specialist school, and the access-to-services basis of the ASGS classification itself (Australian Bureau of Statistics, 2021), together explaining why provision is abundant and large on the coast yet sparse and small inland.

5. Ethical Considerations

Three ethical principles guided this analysis.

The principle of rights protects each individual's dignity and freedom to choose for themselves. In this dataset the custodian suppresses Indigenous and language-background percentages wherever five or fewer students are involved, marking them “np” to prevent the re-identification of individuals in small schools (NSW Department of Education, 2026). Respecting this suppression, treating it as missing rather than reconstructing the values, upholds students' right to privacy; honouring such disclosure controls is fundamental to lawful, trustworthy analytics in industry.

The principle of fairness requires that people and groups be represented in the same way, without distortion or discrimination. Fair encoding choices were made throughout: bars share a common baseline, axes begin at zero, and colour palettes distinguish categories without implying value judgements, guarding against the misleading designs that erode trust in charts (Cairo, 2019). Crucially, retaining Lord Howe Island and the small remote schools keeps low-population communities visible rather than discarding them as outliers, giving every region fair representation.

The principle of the common good holds that shared systems and information should benefit everyone. This dataset is published openly so parents, researchers and planners can all draw on it (NSW Department of Education, 2026). Presenting it with clear titles, labelled axes and cited sources makes the findings interpretable and verifiable, supporting public debate about equitable access to education across NSW.

6. Conclusion

The analysis reveals a public school network shaped overwhelmingly by population geography. School size is strongly right-skewed, and both the number and the size of schools fall steadily

with remoteness, leaving Major Cities clearly dominant on each measure while remote areas remain sparsely served. Metropolitan growth accelerated sharply after the Second World War to overtake an early inner regional lead, whereas remote expansion stayed negligible. Surfaced through matched chart types and careful, privacy-respecting preparation, these patterns provide an evidence base for discussion of equitable resourcing, in particular the challenge of sustaining small schools in remote communities.

References

- Australian Bureau of Statistics. (2021). *Remoteness areas*. Australian Statistical Geography Standard (ASGS) Edition 3, July 2021 - June 2026.
<https://www.abs.gov.au/statistics/standards/australian-statistical-geography-standard-asgs/edition-3-july-2021-june-2026/remoteness-structure/remoteness-areas>
- Cairo, A. (2019). *How charts lie: Getting smarter about visual information*. W. W. Norton.
- NSW Department of Education. (2026). *Master dataset: NSW government school locations and student enrolment numbers*. Data.NSW.
<https://data.nsw.gov.au/data/dataset/nsw-education-nsw-public-schools-master-dataset>
- Tufte, E. R. (2001). *The visual display of quantitative information* (2nd ed.). Graphics Press.
- Wickham, H. (2016). *ggplot2: Elegant graphics for data analysis* (2nd ed.). Springer.